

Marketing Data and Technology



Draw Insights from
Marketing Data



Part One: Setting Goals



Identify Key Business Objectives

Key Business Objective: A defined goal or outcome used to plan the desired direction of your company.

Write at least 3 but no more than 5 business objectives that support your business model. Each objective should be SMART.

1	<i>SMART Key Business Objective 1 (required)</i> <i>Increase total eCommerce revenue by 10% in the next 12 months by optimizing high-performing campaigns.</i>
2	<i>SMART Key Business Objective 2 (required)</i> <i>Increase the number of new customers from 1K to 2K within 6 months through UX and UI optimizations.</i>
3	<i>SMART Key Business Objective 3 (required)</i> <i>Increase the number of returning customers from 13K to 25K within 6 months through PPC.</i>
4	<i>SMART Key Business Objective 4 (optional)</i> <i>Increase the number of monthly transactions from 1K to 2K within 6 months through email and SMS marketing.</i>
5	<i>SMART Key Business Objective 5 (optional)</i> <i>Reduce cart abandonment rate by 10% within 12 months.</i>



Identify Key Performance Indicators

Key Performance Indicator (KPI): A quantifiable metric used to determine how effectively your key business objectives are being met. Ensure that the specific metric is clearly identified.

1	<i>Key Performance Indicator 1 for Key Business Objective 1 (required)</i> <i>Total revenue / ROAS</i>
2	<i>Key Performance Indicator 2 for Key Business Objective 2 (required)</i> <i>Conversion rate</i>
3	<i>Key Performance Indicator 3 for Key Business Objective 3 (required)</i> <i>Returning user rate</i>
4	<i>Key Performance Indicator 4 for Key Business Objective 4 (optional)</i> <i>Total transactions</i>
5	<i>Key Performance Indicator 5 for Key Business Objective 5 (optional)</i> <i>Cart abandonment rate</i>



Part Two: A/B Testing Proposal



A/B Testing Proposal: KPI, Variable, and Hypothesis

KPI used as basis for the A/B test

Conversion rate

Variable that will have an impact on the KPI

Checkout UX

Hypothesis for your A/B Test

If we simplify the purchase process by limiting the numbers of checkout steps and adding a guest checkout option, we will increase the conversion rate.



A/B Testing Proposal: Details and results

Details of the A/B test	
Variations being tested:	<i>Multi-step</i>
	<i>One-page + guest option</i>
User groups:	<i>50/50</i>
Data collection tool:	<i>Google Analytics 4</i>
Length of the test:	<i>1 month</i>

Describe how you would determine the results of the A/B test

I would analyze the conversion rates of both groups within the specified timeframe. To do so, I would generate a report in GA4 comparing both traffic sources. The one with the higher conversion rate would be the winner.



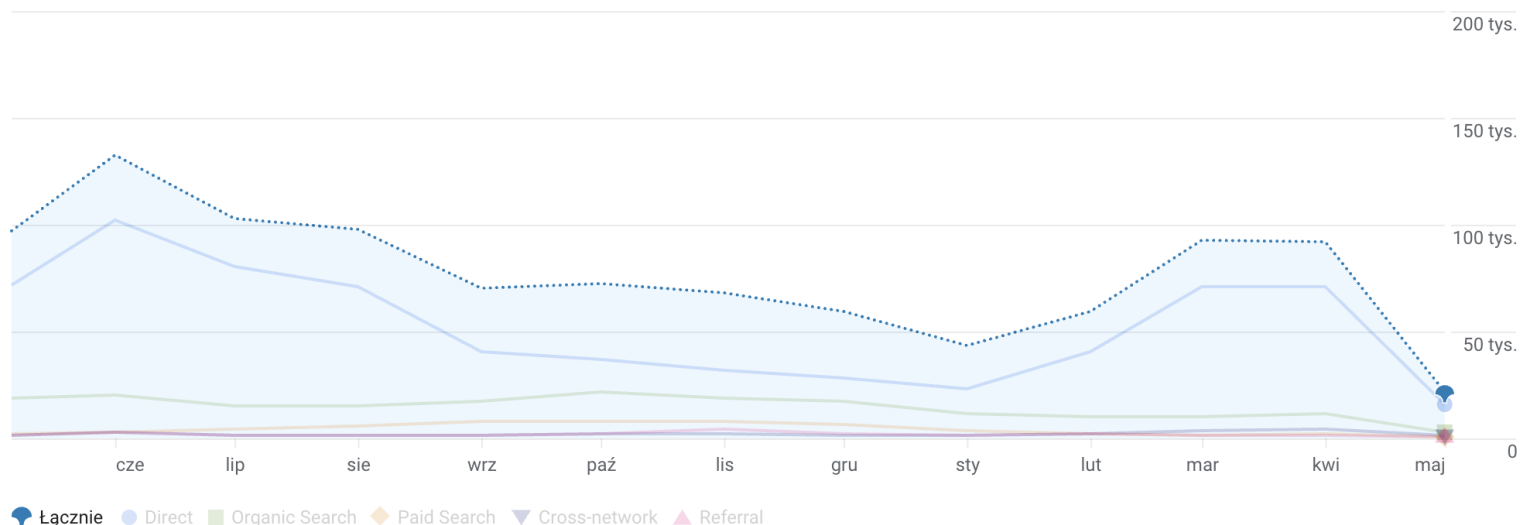
Part Three: Data Exploration



Reports Snapshot

Całkowita liczba użytkowników według wymiaru „Nowy użytkownik – główna grupa kanałów (Domyślna grupa kanałów)” w czasie

Miesiąc ▼





Reports Snapshot

Which month had the most new users?

Jun 2025

Which month had the fewest new users?

Jan 2026

Write some ideas why certain trends are associated with these specific months?

The pattern in your GA4 chart shows a clear seasonal cycle where traffic peaks around June–July and again in March–April, with noticeable drops in late summer, winter, and especially May. This is typical for ecommerce businesses influenced by seasonal demand. The June peak is usually driven by summer-related shopping behavior (travel, fashion, outdoor products) combined with stronger marketing activity before the summer slowdown. The March–April rise often reflects a return to active consumer behavior after winter, supported by spring seasonality, new product launches, and renewed advertising activity. The deep dip in January is also common due to post-holiday fatigue and reduced consumer spending after Christmas and New Year’s Eve.



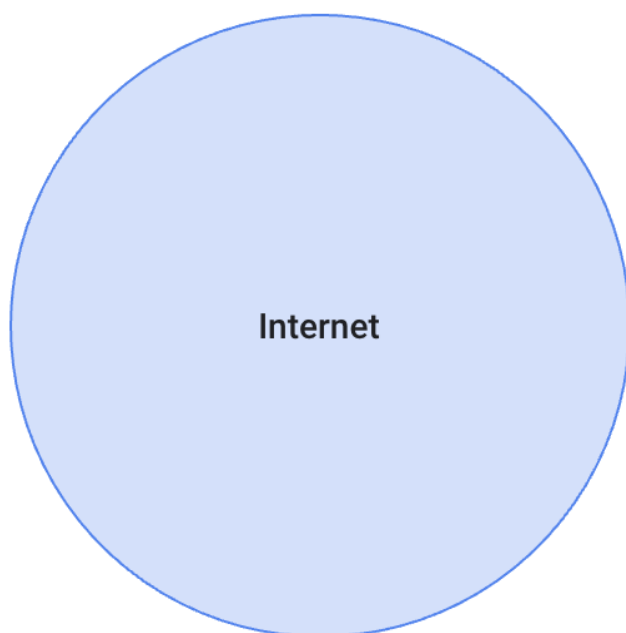
User Tech

Aktywni użytkownicy według: Platforma



WEB

100,0%





User Acquisition



Eksploracja s... ▼



Domyślna grupa kanałów

Całkowita liczba użytkowników

Współczynnik zaangażowania

Łącznie

161 548

94,7%

1

Cross-network

7 319

99,0%

2

Direct

74 769

88,4%

3

Email

6 538

99,6%

4

Organic Search

58 878

98,1%

5

Organic Shopping

1 636

99,0%

6

Organic Social

2 854

99,1%

7

Organic Video

23

97,1%

8

Paid Search

21 400

98,8%

9

Referral

11 095

99,1%

10

Unassigned

2 635

97,1%



User Acquisition

Which channel groups had the highest and lowest engagement rates?

- **Highest: Email (99.6%)**
- **Lowest: Direct (88.4%)**

Which channel groups had the highest and lowest total revenue?

- **Highest: Direct (\$1,281,326.78)**
- **Lowest: Organic Video (\$122.16)**

What do these metrics mean, based on your experience?

Engagement rate shows how actively users interact with your site. Higher values suggest more qualified or interested traffic and indicate stronger user interest and involvement.

Revenue indicates which channels actually drive purchases, so a channel can have high engagement but still contribute little revenue if volume or purchase intent is low.



Monetization



Zakupy e-commerce: Nazwa



Niestandardowe 6 maj 2025–6 maj 2026



Łącznie Nano Banana Sweatshirt Chrome Dino Festive Frost Sweater Google Provence Pullover Hoodie Super G Unisex 1/4 Sporty Zip Google Black Eco Zip Hoodie

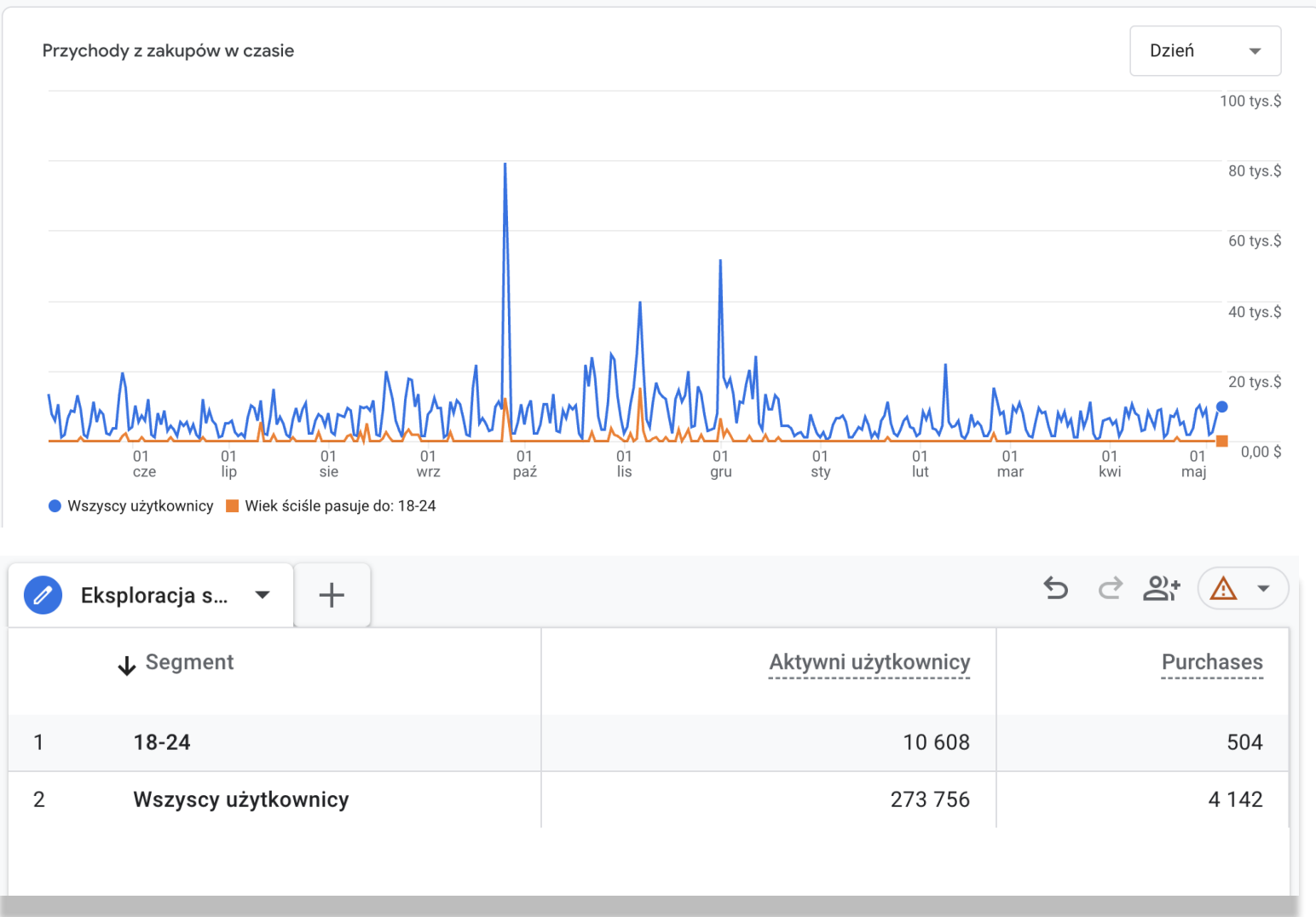
Wiersze wykresu		Szukaj...	Liczba wierszy na stronę: 10		Przejdź do: 1		1–10 z 711	
☐	Nazwa		Wyświetlone produkty	Produkty dodane do koszyka	Kupione produkty	Przychody z produktu		
☐	łącznie		833 626 100% całości	678 177 100% całości	210 522 100% całości	2 771 253,82 \$ 100% całości		
☒	1 Nano Banana Sweatshirt		14 803 (1,78%)	1 973 (0,29%)	507 (0,24%)	33 652,80 \$ (1,21%)		
☐	2 Chrome Dino Festive Frost Sweater		11 399 (1,37%)	4 001 (0,59%)	719 (0,34%)	49 375,20 \$ (1,78%)		
☐	3 Google Provence Pullover Hoodie		11 194 (1,34%)	3 749 (0,55%)	386 (0,18%)	25 608,60 \$ (0,92%)		
☐	4 Super G Unisex 1/4 Sporty Zip		9 136 (1,1%)	951 (0,14%)	274 (0,13%)	15 442,20 \$ (0,56%)		
☐	5 Google Black Eco Zip Hoodie		8 952 (1,07%)	1 675 (0,25%)	484 (0,23%)	27 296,40 \$ (0,98%)		
☐	6 Google Hakone Quilted Jacket		8 825 (1,06%)	1 563 (0,23%)	382 (0,18%)	33 946,00 \$ (1,22%)		
☐	7 Google Driftwood 1/4 Zip		7 679 (0,92%)	1 336 (0,2%)	350 (0,17%)	19 609,80 \$ (0,71%)		
☐	8 Google Kyoto Sweatshirt		7 478 (0,9%)	1 345 (0,2%)	363 (0,17%)	20 617,20 \$ (0,74%)		
☐	9 Google Canyonlands Sweatshirt		7 342 (0,88%)	410 (0,06%)	45 (0,02%)	1 787,10 \$ (0,06%)		
☐	10 Google Arles Sweatshirt		6 978 (0,84%)	1 448 (0,21%)	395 (0,19%)	22 121,40 \$ (0,8%)		



Part Four: Segmentation



Audience Segment: Demographics



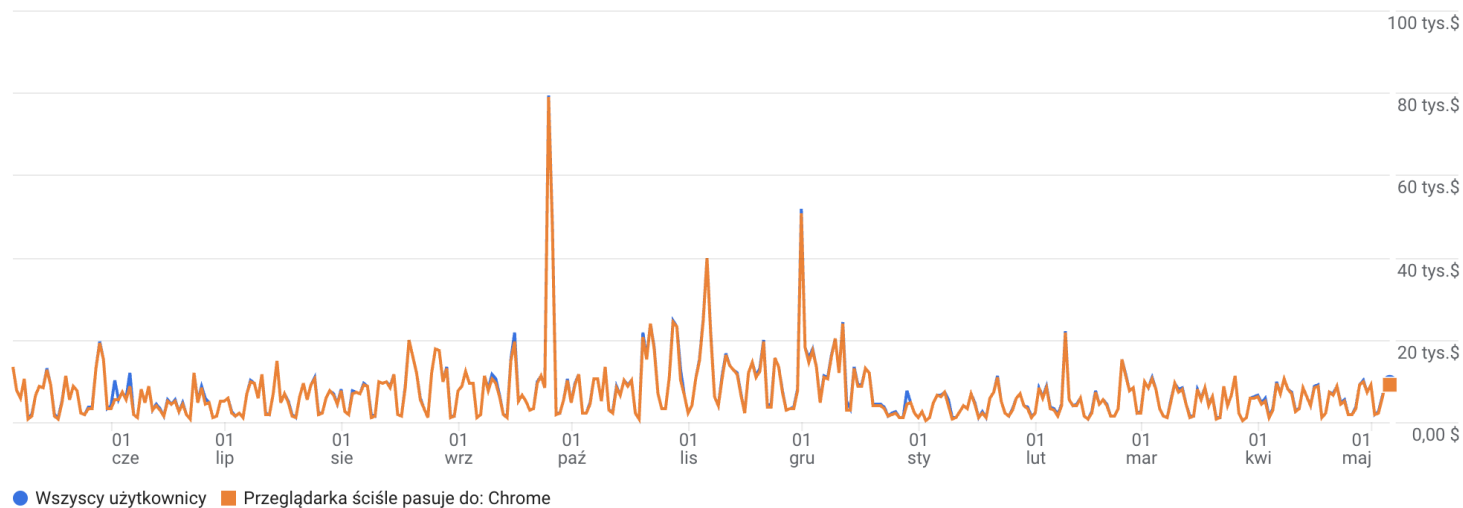
Values used: Age exactly matches 18-24



Audience Segment: Technology

Przychody z zakupów w czasie

Dzień



Eksploracja s...



↓ Segment

Aktywni użytkownicy

Purchases

1	Chrome	245 043	3 909
2	Wszyscy użytkownicy	273 756	4 142

Values used:

Browser exactly matches Chrome



Part Five: Analysis and Suggestions



Google Merchandise Store data

You can find the results of the Google Merchandise Store campaigns below.

Campaign Name	Cost	Revenue	ROAS
Tech Trends: Discover the Latest Google Gear	\$5,000	\$3,000	0.6
Shop with Google: Unleash Your Digital Lifestyle	\$5,000	\$8,000	1.6
Google Gadgets Galore: Elevate Your Tech Game	\$5,000	\$8,000	1.6
Gear Up with Google: Your One-Stop Tech Shop	\$8,000	\$13,000	1.625
Google Merch Madness: Score Big on Tech Essentials	\$5,000	\$2,000	0.4
Unlock the Power of Google: Shop the Best in Tech	\$2,000	\$3,500	1.75
Totals	\$30,000	\$37,500	



Business Sales Growth

To potentially achieve at least 20% YOY sales growth without increasing the current \$30,000 ad spend, the company should reallocate budget away from the lowest-performing campaigns and invest more into the highest-performing ones based on ROAS. Specifically, “Google Merch Madness” (ROAS 0.4) and “Tech Trends” (ROAS 0.6) are significantly underperforming and together consume \$10,000 while producing just \$5,000 in revenue. That budget should instead be shifted toward the top-performing campaigns, especially “Unlock the Power of Google” (ROAS 1.75), “Gear Up with Google” (ROAS 1.625), “Shop with Google” (ROAS 1.6), and “Google Gadgets Galore” (ROAS 1.6). If the reallocated \$10,000 performed at an average ROAS of approximately 1.65, it could generate about \$16,500 in revenue instead of \$5,000. This would increase total projected revenue from \$37,500 to roughly \$49,000. This represents an estimated 30.7% increase in sales, exceeding the 20% YOY growth target without requiring any additional budget.



eCommerce improvements

UX change: Introduce a guest checkout flow as the default path, removing any requirement to create an account before purchase. This would streamline the checkout process by removing barriers at the most critical conversion point, while still offering optional account creation after the order is completed. It improves completion rates, especially for first-time and mobile users, by eliminating one of the most common abandonment triggers.

Other change: Implement a personalized recommendation engine across product pages, cart, and post-purchase flows. This would surface context-aware suggestions such as “frequently bought together,” “complete the set,” or “recommended based on browsing history,” helping users discover relevant items faster and increasing average order value without making the purchasing process more difficult.



Technology

It is time for some exploration! You need to find 2 emerging marketing technologies that you could use in a technology stack. For each one, you need to describe why you would use that tool.

1 *AI-powered Conversational Commerce*

I would utilize AI shopping assistants since they duplicate in-store coaching in a digital setting. These tools can answer product questions, recommend products based on user preferences, let users compare options, and even aid them throughout checkout. This makes it easier for users who are overwhelmed by options or unsure about product details. It also boosts engagement and can recover abandoned carts by proactively communicating with clients throughout their purchasing experience.

2 *Generative AI Marketing Automation Tools*

I would employ generative AI marketing solutions because they greatly accelerate and scale content development and campaign testing. These systems can generate ad copy, email variations, product descriptions, and landing page content, which are then optimized based on performance data. This reduces the time spent on manual content creation while allowing for ongoing A/B testing and customization. As a result, marketing becomes more adaptable, data-driven, and efficient, particularly in rapidly changing e-commerce contexts.